

Lecture 4

Handling Ambiguity

Definition

Ambiguity (*noun*)

1. the quality of being open to more than one interpretation; inexactness.

Why are things ambiguous?

- Poor communication
 - The speaker knows what they mean to say, but the audience doesn't understand
- Uncertainty from the speaker
 - The speaker doesn't know what they want
- Multiple interpretations
 - A business person might want “the most accurate model” – but do they really?

Why does it matter?

- You might waste time doing the wrong thing
 - Or worse, show up at the end having done the wrong thing
- Increased tension / decreased trust between groups
 - If you don't understand what a stakeholder wants, their future feedback might come as a surprise
 - If a stakeholder doesn't understand what you do, they won't know how to interpret and implement your findings
- Missed opportunities
 - If you don't understand well what the issue is, you might miss some possible solutions

Strategies for handling ambiguity

- Clarification
 - Ask questions to gather more information
- Paraphrasing
 - Repeat what someone said back to them
- Confirmation
 - State what you took away from what they said

Clarifications

- Ask clarifying questions

We have a prediction model but it's not working well.
We'd like to replace it with a better one.

Clarifications

- Ask clarifying questions

We have a prediction model but it's not working well.
We'd like to replace it with a better one.

What specifically is not working well in the model?

What metrics are you using to measure how the model is working?

What type of model are you using right now?

Paraphrasing

- Repeat what someone said back to them

We want to forecast demand so that we can plan our schedule more effectively

You want to predict how many customers you will have every hour so you can make sure you have enough staff available at that time

Paraphrasing

- Repeat what someone said back to them

We want to forecast demand so that we can plan our schedule more effectively

Confirmation

- Repeat what someone said back to them

We want to forecast demand so that we can plan our schedule more effectively

You want to predict how many customers you will have every hour so you can make sure you have enough staff available at that time

Confirmation

- Repeat what someone said back to them

We want to forecast demand so that we can plan our
schedule more effectively

Ambiguity due to uncertainty

- Sometimes ambiguity isn't avoidable
 - “What accuracy will your model have” – you don't know until you test it
 - “How long will job X take” – you might not know what it will take to do X
 - “How much data do you need” – there's a lot that goes into this

Ambiguity due to uncertainty

- Sometimes ambiguity isn't avoidable
- When this happens:
 - Acknowledge the uncertainty
 - Provide what information you currently have
 - Share what you don't know
 - Suggest alternatives for different scenarios
 - Set timelines and expectations

Providing ambiguous answers

What accuracy can your model achieve? Can it reach 90%

Providing ambiguous answers

What accuracy can your model achieve? Can it reach 90%

I can't say what accuracy is achievable because every case will be different. If the classes in the data are well differentiated, then the accuracy can be very high. If not, accuracy might be lower. I will look at different modelling and tuning options to get the best result possible.

Providing ambiguous answers

- You are defining a project where you will analyze images and count the number of cars in a photo. How would you answer these questions:
 - How many training images do you need?
 - What model will you use?
 - How long will this take?

Wrapping Up

- Ambiguity can lead to wasted effort, increased conflict, and decreased trust
- Ambiguity can come from several sources, such as:
 - Mismatched communication styles or vocabularies
 - Unknown/uncertain information
- To clarify ambiguous statements, you can:
 - Ask clarifying questions
 - Re-state what you heard
 - Ask for confirmation of your understanding
- If you are asked a question you must give an ambiguous answer to:
 - Acknowledge the source of uncertainty
 - Provide what information you have, and state what you don't know
 - Set timelines and expectations for different situations